

Project Report Submission Sheet

ARIN 350

Responsible AI

Course Project Report: Ethical AI Use Case Development

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Introduction

For my course project, I wanted to explore how artificial intelligence can be used responsibly in a real-world educational setting. As AI tools have become increasingly common in classrooms, I believe it is important to understand both the potential and the ethical risks they introduce. I chose to focus on a K–5 adaptive STEM tutor because early learning is deeply influential. Children are also uniquely vulnerable to algorithmic mistakes at younger ages. This project allowed me to examine the use case from multiple perspectives. I was able to look at industry relevance, bias, fairness, privacy, governance, and generative AI ethics. By analyzing each of these areas, I was able to form a clearer picture of what responsible AI should look like when the system is designed specifically for young learners.

Industry Sector Relevance & Ethical Considerations

I have decided to select the education industry for this project because early learning is one of the areas where AI has the potential to make a meaningful difference in a positive way. It is also where the risks are the highest if it is not designed responsibly. In K–5 classrooms, students learn at very different speeds. Teachers are often responsible for balancing whole class instruction with individual student needs. An AI adaptive learning tutor offers a way to personalize early learning experience without requiring teachers to manually adjust material for each and every student. This is something that is extremely difficult to do consistently during a normal school day.

The AI use case I am proposing is an adaptive learning tutor designed for early STEM education. It will specifically be focusing on math and science reasoning skills. The goal of the system is to tailor instruction to a student's current performance level by adjusting question difficulty, pacing, and lesson sequencing in real time. The AI would also give teachers a dashboard to track trends and identify which students need more help or more challenging work. I chose STEM because early exposure to math and scientific reasoning is strongly connected to long-term learning confidence. It is especially connected with higher grades. This kind of system could give students the support they need before they get left behind in their education.

I think this case is important because it directly supports learning equity. Many students enter school already having been exposed to math games, educational technology, or concept-rich conversations at home. Other students are just beginning to form these skills. An adaptive tutor can help close that gap by giving every student a personalized path. Though, this kind of system also carries risks because young children are considered a legally protected group. Children are covered under government regulations such as COPPA and FERPA. These laws recognize that children cannot fully understand data collection or automated decision-making. This means any AI system interacting with them must avoid mislabeling, over-simplifying, or unintentionally tracking them based on early mistakes. An example would be if a tutor misinterprets a second grader's slow reading of a science problem as a comprehension issue. This interpretation might cause the AI system to lower the difficulty in a way that shapes that child's learning trajectory. This possibility reinforces why responsible design is essential for a tool aimed at early learners.

This project is highly relevant to responsible AI because it involves children's data, educational outcomes, and developmental fairness. Children are a protected group not only in legal terms but also in ethical and developmental terms. This is because children are still developing their understanding, confidence, and learning habits. Global organizations like UNESCO have emphasized that AI in education must support inclusion, transparency, and safety rather than simply optimization or automation (UNESCO, 2021). For my use case, that means building clear mechanisms for teacher oversight, strict data-minimization practices, and age-appropriate explainability. This will ensure that educators and parents understand why the AI is guiding a student in a particular direction. Responsible AI also means the system should enhance, never replace, the teacher's role. Human revisions, checks, and audits are necessary to catch bias, support fairness, and protect children from automated decisions that could shape their

academic identity too early. Learning in early childhood is highly contextual, and a responsible AI system must respect that reality.

By making the key core of this project responsible AI development, I can explore both the promise and the risk of using adaptive tutoring systems in early STEM learning. I chose this use case because it has a clear educational benefit but also raises meaningful questions about fairness, privacy, and governance. The combination of all of these ethical questions makes it a strong foundation for the rest of the course project as I continue to refine how the system should be designed and deployed.

Bias and Fairness

While continuing to build on my earlier proposal, I developed an AI based adaptive STEM tutor for K–5 classrooms to personalize math and science learning. I realized that while personalization seems like a universally positive feature, the underlying algorithms can still introduce subtle forms of bias that influence how children learn and progress. Since these students are young, and their academic identities are still forming, even small algorithmic misinterpretations can have lasting effects.

A major concern for the tutor would be training data bias. If the model is trained primarily on datasets collected from well resourced, English dominant districts, the patterns it learns may not be accurate. This could be reflected by students from diverse linguistic or socioeconomic backgrounds who receive different work based on the bias. For example, a student who pauses before answering may be demonstrating careful reasoning. Though, the system, which is trained on faster response patterns, may misclassify this as a sign of struggle. These misread signals can cause the AI to quietly lower task difficulty. This would reduce the student's opportunities for deeper challenges and potentially harm their confidence over time.

Another risk is feature bias. This occurs when certain performance indicators are weighted more heavily than others. If speed is treated as more important than accuracy, the system may unintentionally reward quick guessers over reflective thinkers. I can imagine two fourth graders solving the same STEM question. The first student blurts out an answer immediately, while the other takes time to check her reasoning. If the AI is not carefully designed it could push the first student ahead and slow the second student down. This is even though the more deliberate learner may have stronger conceptual understanding. Over the course of a year, these repeated micro-biases could create systematic disadvantage for particular types of learners.

I believe to address these issues, bias mitigation must be approached from technical, human, and governance angles at all times. Technically, the system should be trained on diverse and representative data, regularly stress-tested with synthetic cases, and examined through recurring bias audits. Developers should document how different features contribute to adaptive decisions and adjust weights when patterns appear discriminatory. From the human side, teachers should serve as active reviewers. They would have the ability to override recommendations, flag issues, and request reevaluation. Governance practices should require transparency reports, clear documentation, and strict data-handling policies. These practices will help to prevent bias from being reinforced through data collection itself.

For fairness, I believe the goal is to ensure that the system does not systematically advantage or disadvantage any group of students. Fairness means that every child has an equal opportunity to master STEM concepts regardless of language background, processing speed, or previous exposure to academic content. To achieve these goal we will need to include strategies such as teacher guided adjustments, explainable outputs, and intervention workflows. This will be necessary when the AI's recommendations appear inconsistent with a student's true abilities. When educators, parents, and students understand the AI's reasoning, trust increases, and potential inequities become easier to identify and correct (Baker & Hawn, 2022).

Fairness does face several challenges. One challenge is that teachers may not initially feel confident interpreting algorithmic decisions. This could be especially true if the system's explanations are overly technical. Another challenge is that early learners vary widely in communication styles. This means that the system must avoid rigid or simplistic interpretations of behavior. Model drift is an additional concern for the system. Over time as the AI updates, fairness patterns must be re-checked to ensure new biases do not emerge. Addressing these challenges will require ongoing teacher training, age-appropriate explainability, and governance structures that continuously monitor how the system treats different groups of learners.

To combat these challenges, I believe that a proactive, multi-layered approach to bias and fairness must be used. This can make adaptive learning tools genuinely supportive of equity in early education. The key is not to eliminate AI from the process but to ensure that its decisions remain transparent, adjustable, and aligned with each child's right to learn fairly. If we use careful oversight, recurring audits, and strong human involvement, this technology can expand learning opportunities rather than restrict them.

Privacy

I wanted to build on the fairness and bias concerns I discussed earlier. I want to now focus on how privacy and data protection are handled within my AI Adaptive STEM Tutor for K–5 students. Since this system would collect sensitive, educational, data from young children; it needs to have protective, secure, and transparent privacy safeguards at every step. Children’s data is uniquely sensitive because they cannot fully understand digital risks. They also cannot provide informed consent on their own. This makes privacy not only a technical requirement but also an ethical foundation for the entire system. Protecting student information builds trust among parents, teachers, and schools. Securing student information also ensures compliance with legal frameworks like COPPA and FERPA. They are specifically designed to safeguard minors’ information. If there were no strong protections, the technology could unintentionally expose or misuse student data. This could potentially influence how children are taught, evaluated, or even perceived by educators over time.

In this context, privacy means safeguarding all personal and learning data generated by students. It is also ensuring that its collection, storage, and use are transparent and limited to educational purposes. Children’s learning data is deeply personal. This data reflects not only academic performance but also behavioral patterns, mistakes, and progress over time. If there were no proper data handling practices, this information could be exposed or misused. This could potentially lead to the labeling or tracking of children in ways that impact their educational futures. According to Ammanath (2022), trustworthy AI depends on transparent data governance and the principle of informed consent, where users—and in this case, parents—must know how data is gathered, processed, and shared. I believe that same principle should apply to educational technology in every classroom.

The main privacy concerns for my use case include data overcollection, unclear storage policies, and insufficient parental consent mechanisms. An example would be if the adaptive tutor automatically records every keystroke or voice input without parental awareness, it would violate both ethical expectations and federal law. The Children’s Online Privacy Protection Act (COPPA) explicitly requires verifiable parental consent before collecting or sharing personal information from children under 13 (Federal Trade Commission [FTC], 2024). To be able meet this standard the system should allow parents to review and approve the types of data collected. This could be used on data such as performance metrics or engagement analytics. The system should also have the ability to revoke consent at any time. This would help create a safeguard.

In a classroom setting, I imagine a scenario where the AI tutor suggests different problem sets based on each student’s recent activity. The teachers could access dashboards that show patterns in student understanding. These dashboards would give insight into student learning but must avoid exposing identifiable personal information. By using privacy-by-design features, such as anonymized reporting and local data storage, we can help ensure that sensitive data never leaves the school’s secure environment. With transparency reports we could explain, in simple terms, how data is used to adjust lessons. This allows both teachers and parents to have access to these summaries.

In order to strengthen privacy protection, I would implement many different strategies. The different strategies I would include would be encrypted data transmissions, strict access controls, and clear opt-in consent forms for parents. An equally important matter is ensuring that the AI model only retains minimal data necessary for learning insights. It needs to be deleting outdated or irrelevant data on a regular schedule. Additional safeguards should include breach-response protocols, monitoring tools that detect unusual data access, and clear rules preventing the AI from collecting unnecessary information. This could include AI collectors such as facial expressions or emotional indicators. Ammanath (2022) emphasizes that responsible organizations must make transparency “an active practice, not a passive promise,” which I interpret as a commitment to continuous communication with users about how their information is used. There are still several challenges that remain. A major one is balancing personalization with privacy. If you have too little data it limits the system’s accuracy. While, if it has too much it creates significant risk. Another challenge is ensuring that teachers and parents understand privacy settings and can confidently interact with them. I will address these challenges through strong governance structures, ongoing teacher training, and clear, accessible transparency reports. This will help ensure that the system can maintain trust, compliance, and educational value while keeping children’s data safe.

Governance

I want to focus on the bias, fairness, privacy, and ethical concerns I explored in earlier units. With this in mind, I now want to build on how responsible governance can guide the use of my AI Adaptive STEM Tutor in a K–5 setting. This system would be jointly managed by the vendor and the school district to ensure shared accountability and oversight. I believe governance frameworks are essential to make sure the technology stays accurate, safe, and aligned with the best interests of children. Governance should not just be a policy requirement, it should also be the structure that ensures every decision the AI makes is transparent, correctable, and grounded in human oversight.

In my use case, I think the organization should use a combination of the NIST AI Risk Management Framework and the OECD AI Principles. NIST emphasizes practices for mapping risks, measuring impacts, and managing AI behavior through continuous evaluation. I agree with their statement that trustworthy systems require “ongoing monitoring, documentation, and human review” (NIST, 2023). This aligns perfectly with the needs of a K–5 environment where learning accuracy is critical. The OECD principles also stress that AI affecting children must be fair, explainable, and accountable, noting that “responsible stewardship is essential when AI has social impact” (OECD, 2022). Education has long-term developmental consequences, so these frameworks provide a strong foundation for both the vendor’s design responsibilities and the school’s oversight duties.

In terms of actual policy, the partnership should implement clear data access rules, documentation expectations, and a formal oversight workflow. The teachers would serve as the central governors for the system. They would be able to override or pause AI recommendations. This level of classroom control is essential because teachers are the first to notice when something feels off. For example, if the AI suddenly begins recommending repetitive or irrelevant problem sets due to an internal logic error, a teacher should be able to immediately halt the system and report the issue before it affects more students. Without this type of governance, misaligned recommendations could go undetected. This could subtly shift learning paths in ways that contradict earlier fairness and accuracy goals. By giving teachers fast and simple tools for reporting concerns, governance becomes practical rather than theoretical.

Bias has been a key issue in earlier units. I believe bias audits must be the highest priority. They must be conducted at launch and on a recurring schedule. Along with these audits, the organization should also require privacy audits (to confirm COPPA- and FERPA-aligned data practices), security audits (to ensure safe storage of children’s learning data), and performance audits that measure accuracy, learning outcomes, and hallucination rates in generative features. A shared audit dashboard between the school and vendor would maintain transparency and help identify emerging issues earlier, especially if the AI begins drifting away from expected fairness patterns.

In order to support implementation, the organization should adopt strong documentation practices, teacher training sessions, and a clear escalation channel when the AI behaves unpredictably. Potential problems include limited technical expertise in K–5 schools, inconsistent staff training, and the difficulty of keeping governance structures updated as the

system evolves. These challenges can be addressed through continuous professional development, simplified reporting tools for teachers, and formalized communication between the school and the vendor. Ultimately, responsible AI governance ensures not only compliance but also long-term trust. That trust allows the AI to enhance learning while staying aligned with safety, fairness, and transparency.

Mitigation Strategies

Continuing to work on the privacy and fairness issues I explored earlier, I now want to look more closely at the ethical implications of using generative AI within my adaptive K–5 STEM tutor. Generative AI can create explanations, new problem sets, visual diagrams, or even learning pathways for my application. I see it as relevant to the future version of this system. However, with that potential comes a new set of risks. These risks surround the accuracy, safety, and responsible use with young children.

Ethics in this use case means designing generative features that support learning without causing harm, confusion, or misinformation. One of my biggest concerns is accuracy. Generative models can create STEM explanations or examples that sound confident but are mathematically wrong, which could mislead early learners. The Cornell Center for Teaching Innovation warns that generative AI often produces “convincing but incorrect information,” a phrase that highlights how dangerous these errors can be for children who are just forming foundational concepts. If a third grader asks the AI tutor for help with fractions and the model generates a flawed explanation, it could set them back without the student realizing anything is wrong.

Another major ethical issue is over-reliance on AI. Young learners could end up trusting the system more than a human teacher. This is not what we would want. I can imagine a classroom scenario where a student continuously seeks AI-generated hints because they find it easier than thinking through the problem themselves. While the goal is to support learning, generative tools can unintentionally replace productive struggle, a key part of math and science development. Teachers might also become overly dependent on automatically generated lesson materials. This could reduce teacher’s abilities to tailor instructions using their professional judgment.

I wanted to address ethical concerns specifically related to teachers. I believe the system must include clear guidelines and oversight on how educators interact with generative AI. A strong strategy would be to require teachers to review and personalize AI-generated materials instead of relying on them as they are. An example would be a teacher receiving an automatically generated science worksheet. They then before using the material would be prompted to edit, approve, or adjust the content through a required review screen. I can imagine a scenario where a teacher quickly accepts several AI generated assignments out of convenience. An administrative dashboard could flag this pattern and provide reminders or require additional review steps to prevent over reliance on the system. This oversight ensures that teachers remain actively engaged in instructional decisions and continue applying their expertise. This is better than allowing generative tools to take over essential parts of teaching.

There are also important data and consent implications with using generative AI. Since generative AI models often learn from large datasets, it’s essential to ensure that student information is never used to train models in ways that violate COPPA or FERPA. COPPA requires parental consent before collecting or using children's data, and generative systems have

historically blurred boundaries around what data informs outputs. This makes transparency critical. Students and families should always know what information is being used by the AI's generative processes and what safeguards prevent personal data from influencing future model behavior.

I would address these risks by implementing strong accuracy checks, including guardrails that restrict the AI from generating explanations outside validated content libraries. I want teachers to be able to preview, approve, or reject generative materials before students see them. I would also design indicators that help students understand when the AI is making a suggestion rather than presenting a guaranteed fact. I would also make sure model updates are transparent, with clear communication about how data is stored, used, and protected.

The challenge with this application is balancing innovation with safety. Generative AI creates exciting opportunities for customized learning, but only if it remains accurate, supervised, and aligned with ethical standards. If I integrate these tools into my system, they must enhance, not replace, human teaching. This technology must be monitored so that every student continues to learn in a coherent, developmentally appropriate way.

Conclusion

In closing, after completing all sections of this project, I now see how complex and interconnected responsible AI development truly is. This is especially true when it affects children. Bias mitigation, fairness protections, privacy safeguards, governance structures, and generative AI ethics cannot be treated as separate checkboxes. All of these concepts must work together as a single framework that guides every decision the system makes. My K–5 adaptive STEM tutor use case showed me how much care is required to keep the technology transparent, accurate, and aligned with human judgment. I believe with strong oversight, thoughtful design, and collaboration between schools and vendors, AI can support learning without compromising safety or trust. This project strengthened my belief that responsible AI is not a barrier to innovation. Responsible AI makes innovation meaningful, ethical, and beneficial for every student.

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